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Using modified foragers to harvest short-rotation poplar plantations

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ABSTRACT

In Italy, short rotation coppice (SRC) has become very popular in recent years, with over 4000 ha already planted – almost exclusively with clonal poplar. Commercial harvesting operations are dominated by modified foragers, equipped with dedicated SRC headers. At present, contractors use two main header types: German Claas HS-2 and Italian GBE. The former fits smaller foragers with a power below 300 kW, the latter is used on larger harvest units. This study models the performance of modified foragers on a range of short-rotation poplar plantations. Data were collected from 45 operations, covering a total of 118 ha and producing over 4400 green tonnes of wood chips. The average yield of the fields harvested during the trials was about $23 \text{ gt ha}^{-1} \text{ year}^{-1}$ (at a moisture content of 59%), and machine productivity ranged from 9 to 70 green tonnes per scheduled machine hour (gt SMH^{-1}), with an average value of 35 gt SMH^{-1} . A model was developed to predict harvesting performance and cost, showing that harvesting cost can be maintained below the 15 € gt^{-1} (2 € GJ^{-1}) level, if field stocking exceeds 40 gt ha^{-1} .

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1. Introduction

Dedicated crops from surplus agricultural land have the highest potential for increasing the supply of energy biomass, and in the medium term they could account for three quarters of the total supply [1]. Compared to other sources, dedicated crops offer the advantage of an extremely intense management that assures the highest surface yield and the shortest waiting time [2]. As opposed to traditional agricultural crops, biomass plantations can accrue further benefits such as groundwater protection, ecological planning, phyto-remediation etc. [3]. This is especially the case with woody crops, including short rotation coppice [4]. Among various cropping modules, short rotation coppice (SRC) seems to best reflect the expectations of farmers, who are used to short return times and generally show little enthusiasm for traditional wood plantations, harvested at 10–30 years intervals. However, SRC is an industrial crop, designed to produce large quantities of

low-priced raw materials and its success require that all operations be conducted with the utmost efficiency. Harvesting cost is estimated to be above 50% of the total cost of biomass produced from wood plantations [5], which underscores the special needs for optimizing these operations. Within this field, ground-breaking work was initiated 10 years ago by a European working group led by Aberdeen University [6].

At the time, research was mainly focused on willow plantations, as grown in Sweden. This country opened the way by launching an extensive plantation programme in the early 90s, and to date is still the European leader with over 15,000 ha of short rotation willow [7]. Swedish willow plantations are established at a very high density, and harvested every third or fourth year using modified forage harvesters, which have proved very effective [8]. More to the South – in Germany and in Italy, for example – poplar is considered better suited to the local environment than willow, and plantation programmes

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are largely based on this species. Modern poplar hybrids are highly suitable for ex-arable land. Short rotation coppice established with poplar resembles the Swedish model, but it also has some peculiar characteristics that may affect harvesting technology – among other things. Compared to willow, poplar wood is lighter and more brittle, and poplar stools tend to generate fewer and larger sprouts when coppiced [9], which may have a considerable impact on harvesting performance.

In Italy, SRC is very popular in the North, along the Po Valley, where the Italian agricultural industry is concentrated. The Regional Government of Lombardy has been the first one to release grants for the establishment and the management of SRC, and after that several other regions have followed. The result is evident: from 2003 to 2008, over 3000 ha of SRC have been established in Lombardy, representing three quarters of the total SRC in Italy. Plantations in Northern Italy are established almost exclusively with poplar: sites have a good water availability, farmers are already familiar with this tree and specific clones are available for biomass production [10]. Hence the interest in finding the most effective harvesting system, which has led to a number of different technical developments, including forager-based and tractor-based harvesting units. To date, all commercial operations have utilized large-size foragers, equipped with dedicated SRC headers. The first headers were imported from Germany, and were the standard Claas HS-2 developed for the Swedish market. Later on, an Italian manufacturer developed its own beefed-up version, which has equipped all units deployed after 2003 (Table 1).

The goal of this study is: 1) to document and model the performance of modified foragers on a range of short-rotation poplar plantations; 2) to identify technical obstacles to the deployment of these machines and 3) to compare the two main header types used in commercial operations. In the process, a costing model will be developed and applied, for providing unambiguous reference figures.

2. Materials

The study tested five different Claas foragers of the Jaguar Mega series, namely the 840, 850, 860, 880 and 900 models, with engines rated at 254, 286, 306, 340 and 445 kW respectively. These machines were fitted with either the German Claas HS-2 header or with the Italian GBE version. Both units adopt the same structural design, in which a pair of large diameter circular saws is placed at the bottom of the same shafts that

carry the steel-finger crop-collectors. The header cuts the stems and moves them towards the horizontal infeed rollers built into the forager unit, which in turn feeds the stems to the forager chopper. This is the same drum used for chopping maize, but with every other blade removed in order to produce larger wood chips. The main difference between the two headers is that the GBE is heavier and stronger, is fitted with its own additional horizontal infeed roller and uses a hydraulic transmission for most of its functions. In contrast, most of the transmission devices on the Claas header are mechanical. In both cases, the system is based on a forager and 2–4 tractor-trailer units, which receive the chips from the forager and move them to a collection point. The chips are loaded onto transportation vehicles and moved to the energy plant.

The machines were studied while carrying out their scheduled commercial activity, on 45 different sites representative of the two main cropping modules used for SRC poplar in Italy: the annual and the biannual system. Plantations managed on the annual system are harvested at 1-year interval and adopt a planting density of about 10,000 cuttings ha⁻¹. Cuttings are planted in twin-rows, with a spacing of 2.7–2.8 m between twin-rows, 75 cm between the rows forming a pair and 45 cm along the rows. Average stem size at harvest reaches 2–3 cm (cut level), with peaks of 6–8 cm. Seeking a better fibre-to-bark ratio, many farmers resort to the biannual system, where the plantation is harvested at 2–3-years interval, and is accordingly less thick. Cuttings are planted in single rows, with a spacing of 2.8–3.0 m between the rows and 0.5–0.7 m along the rows. This results in a stool density of 6–7000 units ha⁻¹. Average stem size at harvest reaches 10–12 cm at cut level.

3. Methods

The study was designed to evaluate machine productivity and to identify the most significant variables affecting it. The data collection procedure consisted of a set of detailed time-motion studies conducted at the cycle level. In general, detailed time studies are more discriminating than shift-level studies and can detect smaller differences between treatments than shift-level studies can [11].

Cycle times for each machine were defined and split into time elements [12] considered to be typical of the functional process analyzed: this was done with the intent of isolating those parts of a routine that are dependent on one or more external factors in order to enhance the accuracy of the productivity models [13]. All time elements and the related time-motion data were recorded with Husky Hunter® hand-held field computers running Siwork3® time-study software [14].

Output was estimated by measuring the volume of all chip containers produced during each test, and by taking sample containers to a certified weighbridge. Moisture content determination was conducted on samples, collected in sealed bags and weighed fresh and after drying for 48 h at a temperature of 103 °C in a ventilated oven.

Row spacing was measured with a tape, and the length of row harvested for each load with a laser range-finder. This

Table 1 – Foragers equipped for SRC harvesting in Italy.

Make	Model	Header	Year
Claas	Jaguar 840	Claas HS-2	2002
Claas	Jaguar 850	Claas HS-2	2003
Claas	Jaguar 860	Claas HS-2	2003
Claas	Jaguar 880	GBE1	2003
Claas	Jaguar 880	GBE1	2005
Claas	Jaguar 900	GBE1	2006
John Deere	7400	Kemper (prototype)	2008

way, it was also possible to calculate the surface actually harvested at each site.

Tests were conducted from 2004 to 2008 on 6 different machines (all the units currently used in Italian commercial operations) that harvested a total 4470 green tonnes on 118 ha at 45 different sites. The valid time study sessions lasted 153 h.

Data were statistically analyzed with both ANOVA and regression techniques to detect and formalize significant differences and relationships [15]. Operating costs were calculated using the procedures described by Miyata [16], with an estimated annual utilization of 1000 h for the forager and 500 h for the header. The corresponding investment costs are between 200,000 and 300,000 € for the forager, depending on the model and 140,000 € for the header: in both cases the depreciation period was assumed to be 8 years. Repair and maintenance cost were estimated at 70% of depreciation, while labour cost was set at 16 € h⁻¹. Fuel cost was assumed to be 0.90 € L⁻¹ (subsidized fuel for agricultural use). The total costs are inclusive of 20% profit and overheads. Similar assumptions were used for the trailer and the tractor-trailer units. The resulting operating costs are 200–280 and 70–90 € h⁻¹, respectively for the forager and the tractor-trailer unit, depending on the models. Conversion to energy figures was obtained on the assumption that the energy content of dry hardwood equals 18.5 GJ t⁻¹ [17].

4. Results

Table 2 contains the basic statistics for the 45 test fields, demonstrating the large variations in field and stem size, as well as in crop yield. Individual field size ranged from 0.2 to 9 ha, with an average of 2.6 ha, which reflects quite well the fragmentation of the Italian agricultural property. The average butt diameter of all stems in a field varied between less than 2 cm and over 7 cm, and field stocking from 7 to 117 gt ha⁻¹.

$$\text{Power} = \text{Processing Power} + \text{Travel Power}$$

$$= a \times (\text{Crop Throughput, t/min}) + b \times (\text{Travel Speed, km/min})$$

$$= a \times (\text{Linear Crop Density, t/km}) \times (\text{Travel Speed, km/min}) + b \times (\text{Travel Speed, km/min})$$

The average yield was 9 odt ha⁻¹ year⁻¹ with a peak at 19 odt ha⁻¹ year⁻¹: all fields had been established with the new Alasia clones “Pegaso” and “AF2”, and the differences in yield underscore significant differences in site fertility and tending care.

Table 2 – Main statistics for the 45 test fields.

	Mean	Std. Dev.	Std. Error	Minimum	Maximum
Hectares	2.64	2.05	0.31	0.20	9.20
Avg. Diameter	4.39	1.54	0.23	1.58	7.10
gt ha ⁻¹	45.65	24.58	3.66	7.20	116.68
gt ha ⁻¹ year ⁻¹	23.45	10.23	1.52	7.20	47.52
Moisture content	59.75	2.83	0.46	52.66	63.40
odt ha ⁻¹ year ⁻¹	9.41	3.89	0.63	3.00	19.44
Years	1.82	0.49	0.07	1.00	3.00

Table 3 – Main statistics for the productivity of two main unit types (GBE and HS-2).

	Mean	Std. Dev.	Count	Minimum	Maximum
gt SMH ⁻¹ , total	35.63	15.74	45.00	9.23	73.37
gt SMH ⁻¹ , GBE	42.10	13.79	30.00	18.47	73.37
gt SMH ⁻¹ , HS-2	22.68	10.73	15.00	9.23	44.23
gt ha ⁻¹ , total	45.65	24.58	45.00	7.20	116.68
gt ha ⁻¹ , GBE	53.36	23.43	30.00	15.10	116.68
gt ha ⁻¹ , HS-2	30.23	19.47	15.00	7.20	71.90
Avg. Diameter, total	4.39	1.54	45.00	1.58	7.10
Avg. Diameter, GBE	5.19	1.03	30.00	3.50	7.10
Avg. Diameter, HS-2	2.79	1.08	15.00	1.58	4.90

Gross machine productivity ranged from 9 to over 70 green tonnes per scheduled machine hour (gt SMH⁻¹), with an average of 36 gt SMH⁻¹ (Table 3). There was a significant difference between the productivities of the units equipped with the GBE header and those fitted with the Claas header: the former harvested much more productive fields and achieved twice the performance of the latter. All the differences shown in Table 3 are significant to the 0.01 level, and the same accounts for the average power of the prime movers, which is 289 kW for those equipped with the Claas header and 343 kW (18% more) for those mounting the GBE header.

Further work has been devoted to model the performance of these units, by modelling time consumption for the main process stages: processing, turning, delays, preparation and relocation.

For a forager, power is mostly used in processing (severing, chopping and accelerating) the crop. This power consumption is expected to be roughly proportional to the crop throughput. Power is also required to overcome travel resistance: for a given machine and field conditions, this power is approximately proportional to travel speed. These two can be summed:

where linear crop density is the amount of material within the forager's swath per unit length of travel. The power equation can then be rearranged to give:

$$\text{Travel time, min/km} = a \times (\text{Linear Crop Density, t/km}) / \text{Power} + b / \text{Power}$$

While the actual total power used may not equal to the machine's rated power, a more powerful machine has higher throughput capacity and therefore would be expected to travel faster at a given linear crop density. This equation therefore provides a logical general form for the harvesting travel time regression analysis. This was the base for calculating the best model for the time consumption taken by harvesting (i.e. processing + travelling), which is shown in Table 4. This equation is the best fit to the experimental data points obtained from 447 observations, each equal to one loaded trailer. It shows that productivity increases with crop density and forager power.

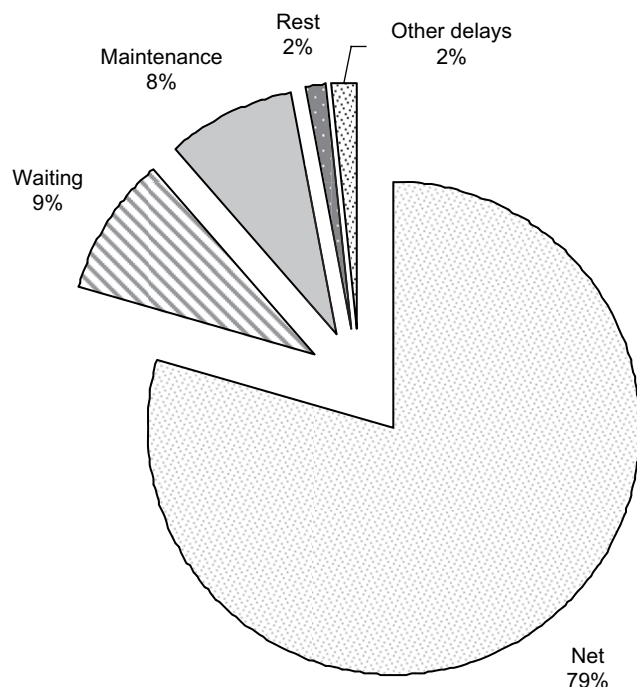
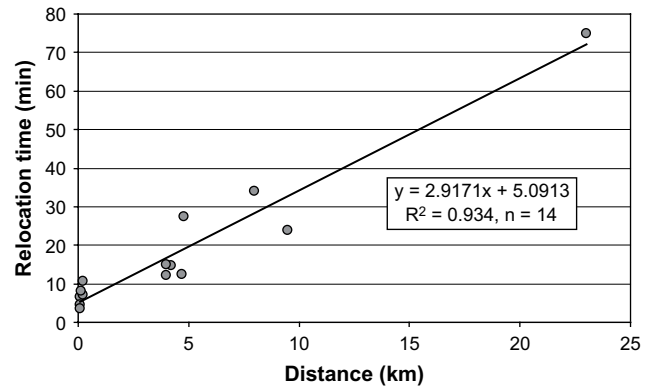
Table 4 – Regression of harvesting time (min/km) vs. crop density and forager power.

	Coefficient	Std. Error	Std. Coeff.	t-Value	P-Value
min/km including blockages vs. 2 independents					
Intercept	–5.322	3.198	–5.322	–1.664	0.0968
kW ^{–1}	3467.189	937.084	0.144	3.700	0.0002
(t km ^{–1}) kW ^{–1}	207.908	12.260	0.660	16.959	<0.0001

Note: R² = 0.385, count = 476.

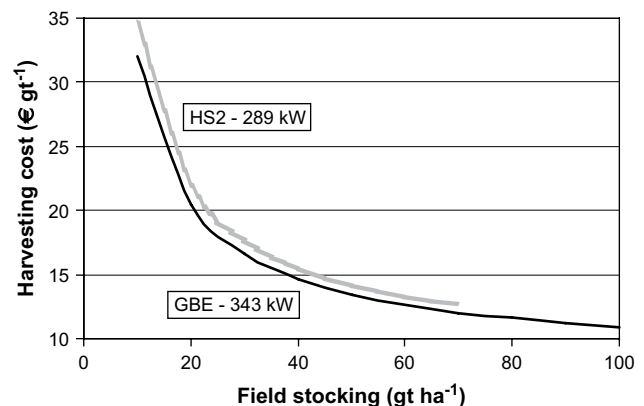
End-row turn times can also be modelled by adopting the median value of 0.55 min per occurrence, calculated from over 2200 turn observations. Standard deviation, minimum and maximum figures were respectively 0.34, 0.05 and 4.39 min.

A good estimate of delay times is then needed to transform net work time (collecting–chipping and turning) into scheduled work time. Delays such as mechanical breakdowns, operator rest and machine interference within the support fleet accounted for less than 21% of the total worksite time (Fig. 1). This figure translates into a 26% average delay factor, which is the coefficient needed to extract delay time from net work time [18]. ANOVA testing could not detect any significant difference between the delay rates experienced by units equipped with the two header types. Similarly, no significant difference was found between the mechanical availabilities of units equipped with different headers, which averaged 92%. Defined as the percent ratio between maintenance-free work time and total work time, machine availability can give a measure of how the harvester copes with the strain of handling wood, rather than softer forage or maize. During the study we measured at 52 min the average duration of daily preparation time (refuel, warm-up, cleaning etc.): standard deviation, minimum and

**Fig. 1 – Breakdown of worksite time (excl. preparation and relocation).****Fig. 2 – Relocation time as a function of distance between fields.**

maximum figures were respectively 12, 25 and 64 min. Finally, relocating the operation could also be modelled: the graph in Fig. 2 shows the relationship between relocation time and distance between the sites, and includes terminal times (respectively “prepare for travel” and “prepare for work”).

The relationships shown above allowed building a simple deterministic model to predict harvesting productivity and cost as a function of: crop density, row length, machine power and expected level of delays. Based on crop density and machine power, the model first calculates harvesting time with the equation in Table 4. It then adds turning time, after calculating the number of turns on the basis of row length and field surface. Delays are then calculated by applying an appropriate delay coefficient to net work time, i.e. harvesting and turning time. The delay coefficient can be typed by the user, but in the absence of a specific user input the model will take the estimate produced by this study (0.26) as a default value. Total worksite time is finally obtained by adding preparation time to total work time (net work time + delays). Finally, relocation time is added to the total, using the function shown in Fig. 2. The model also calculates transport fleet balance and operation relocating time: it is available for free, and can be requested to the authors at: spinelli@ivalsa.cnr.it. Fig. 3 shows the result of a simple simulation, calculated for a forwarding distance of 2 km, a relocation distance of 10 km and an average lot size of

**Fig. 3 – Harvesting cost as a function of field stocking and machine power.**

3 ha. Two harvesters were simulated: a 289 kW forager equipped with a Claas HS-2 header, and a 343 kW forager equipped with the GBE header. Depending on field stocking and machine type, the total harvesting cost including forwarding to a collection site, preparation and relocation range from 11 to 35 € gt⁻¹ (1.5–4.7 € GJ⁻¹). Harvesting cost is restrained below the 15 € gt⁻¹ (2 € GJ⁻¹) ceiling only if field stocking exceeds approximately 40 gt ha⁻¹ and the more powerful unit is used. Although more expensive to operate, the more powerful unit is more productive and offers a lower harvesting cost. Besides, smaller units fitted with the HS-2 heads are seldom used on fields with more than 70 gt ha⁻¹, where stronger foragers are preferably deployed.

5. Discussion and conclusions

The average yield of the fields harvested during the trials was about 9 odt ha⁻¹ and approached the top yield measured further north for the best poplar [19,20] and willow [21] clones. Moreover, the best fields encountered in the study reached annual yields higher than 15 odt ha⁻¹, a witness to the good quality of the Italian clones and climate. We should also note that these values are net, after the harvesting losses.

Italian contractors have developed a very effective harvesting system, based on modified foragers. The current trend is towards powerful units fitted with the heavier and stronger GBE header, which are best suited to plantations harvested every second year: older, less powerful units equipped with the HS-2 header were originally meant to harvest less-developed annual rotations, and they are actually used on both annual and biannual crops, on condition that stems are not too big. In our sample, the richer field harvested by these units had a stocking of 70 gt ha⁻¹, which is then assumed as the upper limit. However, the effective harvesting of large-size stems requires more than just a powerful forager: plantation design must adapt by allowing for an accordingly large spacing, so that the tops of cut stems can sneak between the standing crop ahead and the stems can be laid horizontal. Modified foragers cannot harvest stems that are too big and too close.

Top performance is obtained only when several factors concur, and namely: good terrain conditions, adequate machine choice, high crop density and appropriate row spacing. A forager is a heavy machine that cannot traffic wet or sloping soils, and therefore should only be applied to flat and solid terrain. If harvesting conditions are appropriate, modified foragers can reach a very high productivity, with peak values up to 70 gt h⁻¹, including turns and delays.

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